Raj Golakiya

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EDUCATION

Lambton College In Toronto, ON Jan'23 – Aug'25

Artificial Intelligence and Machine Learning

- Investigated data science, Python Programming, big data basics, Ai career pathways, professional communication, and applied AI principle to practical problems.
- Examined data visualization techniques and tools, advanced python for AI/ML, data mining concepts, Natural Language Processing Fundamentals, and effective management of AI/ML projects.
- Studied neural network basics and practical applications, explored social media analytics, learned essential software tools for AI/ML, and complete a collaborative AI/ML capstone project with real-world application and competition.

Vivekanand College, Surat, Gujarat 2019 – 2021

Bachelor Of Computer Application

- Established foundational skills in communication, mathematics, and computer basics during the initial semester.
- Developed proficiency in diverse programming languages such as C, Java, PHP, ASP.Net, and undertook advanced projects, including a thesis where developed a website using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Explored advanced topics like computer graphics, network technology, e-commerce, and cyber security, enhancing practical application capabilities.
- Culminated with a comprehensive understanding of computer applications, incorporating software engineering, system management, and hands-on project development with technologies spanning UNIX, Shell Programming, PHP, MySQL, ASP .Net, and the MERN stack.

PROJECTS

IntelliExam: Intelligent Exam Paper Generator

- **User Input:** Designed a user-friendly interface where users can either provide a topic name or input their own content. Options include selecting the number of questions (10, 20, or 30) and the complexity level (Easy, Medium, Hard).
- **Model Development**: Evaluated multiple models, ultimately selecting GPT-3.5 from OpenAI for its superior performance in generating high-quality, customizable questions. Previous models like LLaMA3 and custom-built models were tested but found inadequate due to biases, limited capabilities, and high time complexity.
- **Question Generation**: Utilized the GPT-3.5 model to generate questions, specifying the desired number and complexity. Questions were formatted into a JSON response, then converted into a downloadable Microsoft Word document using python-docx.
- **Challenges and Solutions**: Overcame challenges in generating related topics and gathering comprehensive text corpora by implementing a combination of API calls and NLP techniques. Addressed model limitations by iterating on different solutions and integrating user feedback for continuous improvement.
- **Outcome**: Successfully created a scalable, efficient, and user-friendly tool that saves educators significant time and effort while providing high-quality, customizable exam papers. Received positive user feedback and identified areas for future enhancements.

Data Science Project *Lambton College*

- **Machine Learning & Statistical Modeling:** Developed and optimized supervised and unsupervised models including Linear Regression, Logistic Regression, Ridge, Lasso, Decision Trees, Random Forest, KNN, SVM, Naive Bayes, K-Means, and XGBoost (classification & regression) for predictive analytics and customer behavior modeling.
- **Deep Learning & Neural Networks:** Designed and trained custom neural networks and leveraged pretrained models for image classification and pattern recognition tasks, working with both structured and unstructured data.
- **Natural Language Processing (NLP):** Built end-to-end NLP pipelines including text preprocessing, normalization, tokenization, and vectorization (Bag of Words, TF-IDF, Word2Vec, and sentence embeddings), applying both extractive and abstractive modeling techniques with custom and pretrained architectures.
- **Advanced Exploratory Data Analysis (EDA):** Conducted in-depth EDA to identify patterns, trends, correlations, and anomalies, improving feature engineering and model performance.
- **End-to-End Data Pipeline Development:** Engineered scalable workflows integrating data collection, cleaning, transformation, and preparation for machine learning systems in both batch and real-time environments.
- **Automation & ML Pipelines:** Built automated pipelines (live data scraping → preprocessing → model training → output updates) to enable continuous learning and dynamic system updates.
- **Data Engineering & Web Scraping:** Extracted and processed data from complex, dynamic platforms (e.g., social media and e-commerce sites) using Selenium, BeautifulSoup, and Chromium, handling anti-scraping mechanisms and dynamic content.
- **Big Data Analytics & Visualization:** Developed interactive dashboards using Tableau and Power BI to present insights from large datasets and support decision-making.
- **Collaboration & Documentation:** Collaborated in team environments, documenting methodologies, data workflows, and model results to ensure reproducibility and knowledge sharing.

TECHNICAL SKILLS & OTHER

- **Programming Languages**: Python, JavaScript
- **Machine Learning Frameworks**: Scikit-Learn, TensorFlow, Keras, PyTorch
- **Data Manipulation and Analysis:** NumPy, Pandas, SciPy
- **Natural Language Processing**: NLTK, SpaCy, Transformers, Beautiful Soup
- **Web Development**: HTML, CSS, Flask, ReactJS, NodeJS, ExpressJS
- **Databases**: SQL, MongoDB, PostgreSQL
- **Development Tools**: Git, GitHub, Jupyter Notebooks, Docker, Goole Colab, Kaggle
- **Additional Skills**: Data Analytics, Data Visualization (Tableau, PowerBI, Matplotlib, Seaborn), Supervised & Unsupervised Learning Algorithms, EDA, Feature Engineering, Feature Selection, Class Balancing, Feature Extraction, Algebra, Probability, tatistics, Calculus, Matrices, and typing speed 70 WPM.